



Universidad
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Universidad Nacional de Loja

Faculty of Energy, Industry, and
Non-Renewable Natural Resources

Electrical Engineering Program

Analysis and Protection of Electrical Substations Using ETAP

Transmission Lines and Electrical Substations

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Assignment Statement

The power system shown in Fig. 1 comprises three transformer substations:

- Substation 1 supplies a predominantly residential load and will be installed in a densely populated urban area with high land costs.
- Substation 2 supplies a predominantly general-industrial load and will be installed on the outskirts of the city, where land costs are moderate.
- Substation 3 supplies a predominantly mining load and will be installed in a rural area with low land costs.

The required engineering tasks are as follows:

1. Select the busbar arrangement for each substation. Define weighting factors for reliability, footprint, maintainability, cost, expandability, operational flexibility, ease of operation, and constructability. Select the preferred arrangement using technically substantiated criteria and the spreadsheet evaluated in class.
2. Design the grounding grid according to the site soil conditions and available area. Develop the design in ETAP and verify it with the spreadsheet supplied in class. The design shall comply with IEEE Std 80-2013. If ground rods are required, select them from actual manufacturers' catalogs. The site data are provided in Table I.
3. Specify the surge-arrester model required for each substation bus, assuming a surge impedance of $300\ \Omega$.
4. For the parameters listed in Table II, determine the spacing of air terminals required to provide adequate lightning shielding. If overhead shield wires are used, determine their required separation.

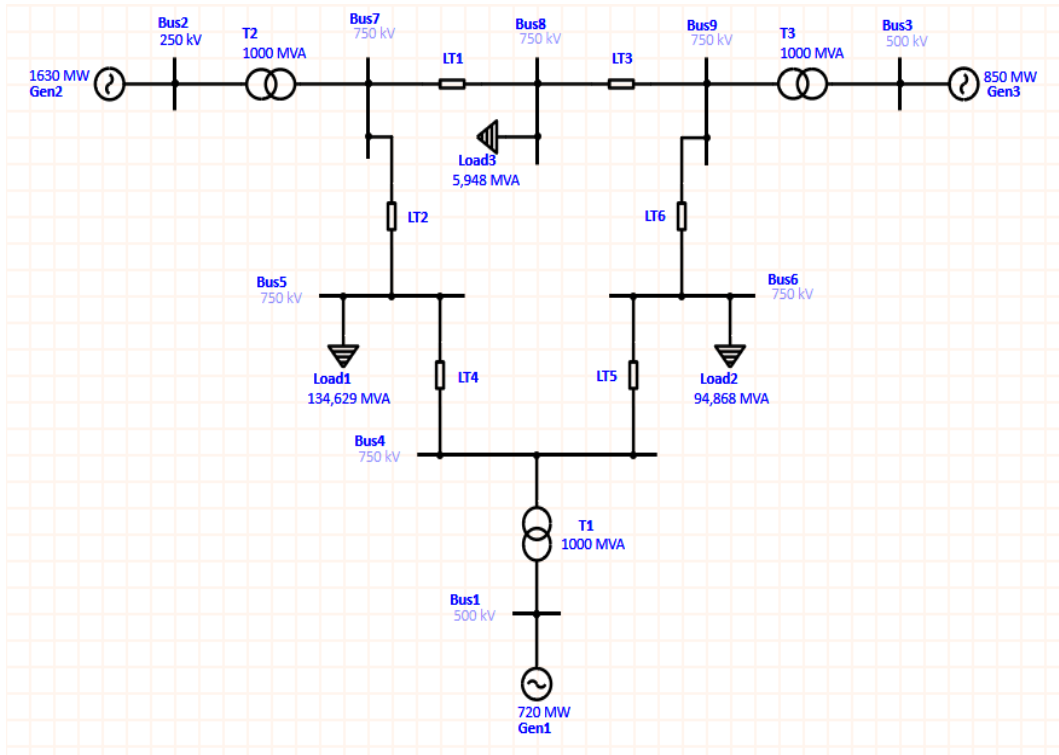


Fig. 1. Electric power system under study.

TABLE I
SITE AND SOIL PARAMETERS

Substation	ρ_s	Surface layer	ρ	Maximum area	Grid depth
S1	2500 $\Omega \cdot \text{m}$	0.25 m	269.32 $\Omega \cdot \text{m}$	$60 \times 60 \text{ m}^2$	0.5 m
S2	2500 $\Omega \cdot \text{m}$	0.25 m	305.5 $\Omega \cdot \text{m}$	$70 \times 40 \text{ m}^2$	0.5 m
S3	2500 $\Omega \cdot \text{m}$	0.25 m	324.3 $\Omega \cdot \text{m}$	$70 \times 70 \text{ m}^2$	0.5 m

TABLE II
LIGHTNING-SHIELDING DESIGN PARAMETERS

Substation	BIL	Terminal/shield-wire height	Equipment height
S1	450 kV	20.5 m	5.5 m
S2	325 kV	18.5 m	5.5 m
S3	350 kV	18.5 m	5.5 m

1. Busbar Arrangement Selection

According to Sivanagaraju [1], an electrical substation is an assembly of equipment used for switching, voltage transformation, power-factor correction, and power or frequency conversion. A transmission or subtransmission substation receives electric power at high voltage, transforms it to a lower voltage level, and supplies primary-voltage feeders for subsequent distribution.

Substations may be classified by function as follows:

1. Transformer substations, which transfer power between voltage levels and may serve transmission, subtransmission, or distribution networks.
2. Industrial substations, which supply large concentrated loads.
3. Switching substations, which interconnect and sectionalize transmission circuits without changing the voltage level.
4. Synchronous-compensator substations, which incorporate synchronous condensers to regulate reactive power and improve system power factor.
5. Frequency-conversion substations, which supply loads requiring a frequency different from the network frequency.
6. Converter substations, which perform ac–dc conversion, particularly in high-voltage direct-current systems.

Substations may also be classified by physical construction:

- Indoor substations, in which equipment is enclosed to limit exposure to contamination and environmental conditions.
- Outdoor substations, in which equipment is installed in open air and a larger equipment footprint is generally available.
- Underground substations, typically used in densely populated areas where surface space is constrained and construction costs are high.
- Pole-mounted substations, commonly used for distribution transformers installed on overhead structures.
- Gas-insulated substations (GIS), in which high-voltage components are enclosed in grounded metal compartments and insulated with gas, providing a compact and highly protected installation.

The busbar system is one of the most critical substation components because it establishes the electrical interconnection of incoming and outgoing circuits. Common arrangements include single bus, sectionalized single bus, double bus, main-and-transfer bus, double bus with double breaker, ring bus, and breaker-and-a-half schemes. This study does not

provide a detailed description of each topology; instead, it applies a weighted decision method to identify the most suitable alternative.

The analysis addresses Substation 3, located between Bus 2 (250 kV) and Bus 7 (750 kV). The substation supplies a mining load in a rural area where land is readily available and comparatively inexpensive. Consequently, supply reliability and operational flexibility are the dominant selection criteria.

The selection is performed using a weighted decision matrix. Each criterion is assigned an importance score from 1 to 5, where 5 denotes the highest priority. The assigned weights reflect the substation duty, the criticality of the connected load, and the site constraints. Table III summarizes the adopted weighting factors.

TABLE III
BUSBAR-SELECTION WEIGHTING MATRIX

Criterion	Weight	Normalized weight
Reliability	5	19%
Physical footprint	2	8%
Maintainability	3	12%
Cost	2	8%
Expandability	3	12%
Operational flexibility	5	19%
Ease of operation	3	12%
Constructability	3	12%

The engineering basis for these weights is summarized below:

- **Reliability (5):** Mining facilities typically operate continuously and impose high power demand. An interruption may cause substantial production losses, equipment damage, and operational safety hazards. Fault tolerance, redundancy, and continuity of supply therefore receive the highest priority.
- **Physical footprint (2):** The rural site provides sufficient low-cost land. Arrangements with a larger footprint can therefore be considered when they offer improved redundancy or expansion capability.
- **Maintainability (3):** The selected arrangement should permit preventive and corrective maintenance without de-energizing the critical load. This requirement is important, although secondary to overall reliability and operational flexibility.
- **Cost (2):** Capital cost remains relevant; however, continuity of service and technical performance take precedence. Low land cost also permits a greater share of the project budget to be allocated to primary equipment and redundancy.
- **Expandability (3):** Future growth in mining activity may require additional feed-

ers, transformers, or generation interfaces. The initial layout should therefore accommodate staged expansion without major reconstruction.

- **Operational flexibility (5):** The arrangement must support system reconfiguration during contingencies, switching operations, and maintenance while preserving supply to the mining process.
- **Ease of operation (3):** Switching sequences must remain clear, safe, and resistant to human error despite the use of a comparatively sophisticated bus arrangement.
- **Constructability (3):** The selected topology must be practical to install and commission while meeting the required technical performance and project constraints.

Each candidate arrangement was rated from very low to very high for reliability, cost, flexibility, and footprint, and from very complex to very simple for maintenance, expansion, operation, and construction. The completed decision matrix is shown in Fig. 2. The breaker-and-a-half arrangement obtained the highest aggregate score, followed by the double-bus arrangement. The breaker-and-a-half scheme is therefore selected because it provides the reliability, switching flexibility, and maintenance continuity required for the critical mining load.

PARÁMETROS		PESO (%)		BREAKER Y MEDIO		ANILLO COMBINADO		DOBLE BARRA		B. dual y transf.	
				Calificación	Puntaje	Calificación	Puntaje	Calificación	Puntaje	Calificación	Puntaje
Confiabilidad	5	19%	Muy Baja	1	0		0		0		0
			Baja	2	0		0		0	X	10
			Media	3	0		0	X	15		0
			Alta	4	0	X	20		0		0
			Muy Alta	5	X	25	0		0		0
Espacio Físico	2	8%	Muy Grande	1	0		0		0		0
			Grande	2	X	4	0		0		0
			Medio	3	0		0	X	6	X	6
			Bajo	4	0	X	8		0		0
			Muy Bajo	5	0		0		0		0
Facilidad de Mantenimiento	3	12%	Muy Complejo	1	0		0		0		0
			Complejo	2	0	X	6		0	X	6
			Moderado	3	0		0	X	9		0
			Sencillo	4	X	12	0		0		0
			Muy Sencillo	5	0		0		0		0
Costo	2	8%	Muy Alto	1	X	2	0		0		0
			Alto	2	0	X	4		0		0
			Medio	3	0		0		0	X	6
			Bajo	4	0		0	X	8		0
			Muy Bajo	5	0		0		0		0
Facilidad de Expansión	3	12%	Muy Complejo	1	0		0		0		0
			Complejo	2	0		0		0	X	6
			Moderado	3	0	X	9		0		0
			Sencillo	4	X	12	0		0		0
			Muy Sencillo	5	0		0	X	15		0
Flexibilidad y Modulación	5	19%	Muy Baja	1	0		0		0		0
			Baja	2	0		0		0		0
			Media	3	0		0		0	X	15
			Alta	4	0	X	20	X	20		0
			Muy Alta	5	X	25	0		0		0
Facilidad de Operación	3	12%	Muy Complejo	1	0		0		0		0
			Complejo	2	0		0		0		0
			Moderado	3	0	X	9		0		0
			Sencillo	4	X	12	0		0		0
			Muy Sencillo	5	0		0	X	15	X	15
Facilidad de Construcción	3	12%	Muy Complejo	1	0		0		0		0
			Complejo	2	X	6	0		0		0
			Moderado	3	0	X	9	X	9		0
			Sencillo	4	0		0		0	X	12
			Muy Sencillo	5	0		0		0		0

TOTAL PUNTOS

26,00	100,00%
-------	---------

 98,00 - 85,00 - 97,00 - 76,00

Mayor Puntuación

98

 BREAKER Y MEDIO

Fig. 2. Weighted evaluation of the candidate busbar arrangements.

2. Substation Grounding Grid

The grounding-grid design follows the procedure established in IEEE Std 80, *IEEE Guide for Safety in AC Substation Grounding* [2]. The principal design inputs for Substation 3, including the maximum fault current obtained from the ETAP short-circuit study [3], are as follows:

- fault duration: 0.5 s;
- current-division factor, S_f : 60%;
- native-soil resistivity, ρ : 324.3 $\Omega \cdot \text{m}$;
- surface-layer resistivity, ρ_s : 2500 $\Omega \cdot \text{m}$;
- surface-layer thickness: 0.25 m;
- grid burial depth: 0.5 m;
- system X/R ratio: 15.13; and
- maximum asymmetrical fault current: 8.799 kA.

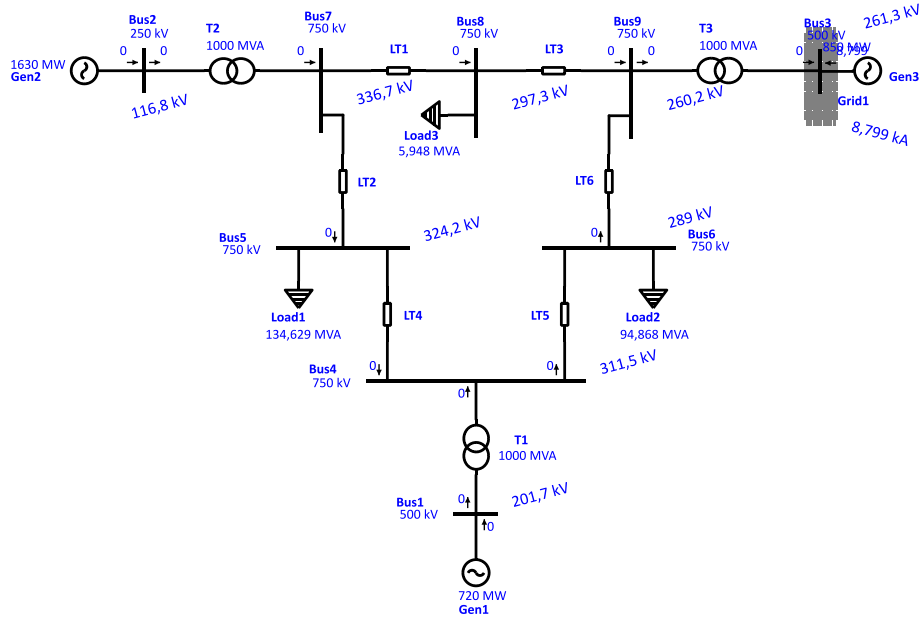


Fig. 3. Power-system model with a single-line-to-ground fault at Bus 3.

The image shows the 'GRD Study Case Editor' dialog box. It contains several sections: 'Study Case' with a text field for 'Study Case ID' (GRD1); 'Reports & Plots' with checkboxes for 'Auto Display Summary & Alert' (checked) and 'Report Details' (unchecked); 'Options' with radio buttons for 'Weight' (50 kg and 70 kg, with 70 kg selected), a text field for 'Ambient Temperature' (40 °C), and an 'Update' section with a checkbox for '# of Conductors and Rods (Optimization)' (unchecked); 'Method' with radio buttons for 'Finite Element', 'IEEE 80 - 2000/2013' (selected), 'IEEE 80 - 1986', and 'IEEE 665 - 1995'; 'Fault Durations' with text fields for 'tf' (0.5 Sec), 'tc' (0.5 Sec), and 'ts' (0.5 Sec); 'Ground Short-Circuit Current' with radio buttons for 'User Specified' (unchecked) and 'Short-Circuit Study' (checked), and text fields for 'Ifg' (8.799 kA) and 'X/R' (15.13); 'Grid Current Factors' with text fields for 'St' (60 %) and 'Cp' (100 %); and a 'Remarks 2nd line' text area. At the bottom, there are navigation buttons (< and >), a dropdown menu (GRD1), and 'Help', 'OK', and 'Cancel' buttons.

Fig. 4. ETAP grounding-grid study parameters.

The maximum short-circuit current and X/R ratio are documented in Figs. 3 and 4. The grounding conductor is then sized for thermal withstand. IEEE Std 80 expresses the required cross-sectional area in kcmil as

$$A_{\text{kcmil}} = I \frac{197.4}{\sqrt{\left(\frac{T_{\text{cap}}}{t_c \alpha \rho_r}\right) \ln\left(\frac{K_0 + T_m}{K_0 + T_a}\right)}}, \quad (1)$$

where A_{kcmil} is the conductor cross-sectional area, I is the asymmetrical rms fault current in kiloamperes, T_m is the maximum allowable conductor temperature, T_a is the ambient temperature, α is the thermal coefficient of resistivity, K_0 is the thermal resistivity coefficient at 0°C , ρ_r is the conductor resistivity at reference temperature T_r , t_c is the fault-current duration, and T_{cap} is the material thermal-capacity factor specified by IEEE Std 80. The guide also recommends a minimum conductor size of 2/0 AWG for mechanical robustness and a nominal ground-rod diameter of 5/8 in.

Description	Material ^a conductivity (% IACS)	α_r factor ^a at 20 °C (1/°C)	K_o at 0 °C (0°C)	Fusing ^a temperature T_m (°C)	Resistivity ^a at 20 °C ρ_r ($\mu\Omega\cdot\text{cm}$)	Thermal ^a capacity $TCAP$ [J/(cm ³ · °C)]
Copper, annealed soft-drawn	100.0	0.003 93	234	1083	1.72	3.4
Copper, commercial hard-drawn	97.0	0.003 81	242	1084	1.78	3.4
Copper-clad steel wire	40.0	0.003 78	245	1084 ^c	4.40	3.8
Copper-clad steel wire	30.0	0.003 78	245	1084 ^c	5.86	3.8
Copper-clad steel rod	17.0	0.003 78	245	1084 ^c	10.1	3.8
Aluminum-clad steel wire	20.3	0.00360	258	657	8.48	3.561
Steel, 1020	10.8 ^b	0.003 77	245	1510	15.90	3.8
Stainless-clad steel rod ^c	9.8	0.003 77	245	1400 ^c	17.50	4.4
Zinc-coated steel rod	8.6	0.003 20	293	419 ^c	20.10	3.9
Stainless steel, 304	2.4	0.001 30	749	1400	72.00	4.0

Fig. 5. Material constants used for grounding-conductor selection.

For commercial-grade copper and an ambient temperature of 40 °C, (1) gives

$$\begin{aligned}
A_{\text{kcmil}} &= 8.799 \frac{197.4}{\sqrt{\left(\frac{3.4}{0.5(0.00381)(1.78)}\right) \ln\left(\frac{242+1084}{242+40}\right)}} \\
&= 44.087 \text{ kcmil}, \\
A_{\text{mm}^2} &= 44.087(0.5056010701) = 22.29045 \text{ mm}^2.
\end{aligned}$$

The calculated thermal requirement is approximately equivalent to a 4-AWG copper conductor, as indicated in Fig. 6. Nevertheless, the mechanically robust minimum size recommended by the design guide is adopted. The selected grid conductor is therefore 2/0 AWG copper, with a nominal cross-sectional area of approximately 70 mm².

AWG/kcmil	MM2
#22 awg	0.5
#20 awg	0.75
#18 awg	1
#16 awg	1.5
#14 awg	2.5
#12 awg	4
#10 awg	6
#8 awg	10
#6 awg	16
#4 awg	25
#2 awg	35
#1 awg	50
#2/0 awg	70
#3/0 awg	95
#4/0 awg	120
300 kcmil	150
350 kcmil	185
500 kcmil	240
750 kcmil	400

Fig. 6. AWG conductor-size conversion data.

The available site area is $70 \times 70 \text{ m}^2$. A $64 \times 64 \text{ m}^2$ square grid was selected, consisting of 25 parallel conductors in each orthogonal direction. The resulting center-to-center conductor spacing is approximately 2.667 m. The catalog unit price of the selected 2/0-AWG bare copper conductor is USD 9.30/m [4]; the corresponding catalog information is reproduced in Fig. 7.



CABLE DESNUDO DE COBRE # 2/0 AWG
COD: C21260

SUAVE RECOCIDO

\$ 9,30

Lo sentimos al momento no contamos con el ítem que estabas buscando
 ☹️. Envíanos un mensaje a: ecommerce@ingelcom.com.ec

[Términos y condiciones](#)
 Garantía de devolución de 30 días
 Envío: 2-3 días laborales

Referencia Interna: C21260

Fig. 7. Catalog data for the selected 2/0-AWG copper conductor.

Ground rods improve current dissipation into the soil and reduce local potential gradients. The adopted design uses 44 copper-bonded steel rods, each $5/8$ in in diameter and 2.4 m long. The selected commercial rod, shown in Fig. 8, has a unit price of USD 11.22.



Fig. 8. Selected copper-bonded grounding rod.

The 64 m grid dimension divided into 24 intervals yields a conductor spacing of 2.67 m, which exceeds the 2.4 m rod length. Although the conductor intersections provide 625 potential rod locations, rods are not required at every node. Parametric ETAP simulations established that 44 rods provide acceptable touch- and step-voltage performance with lower material cost. The final arrangement is shown in Fig. 9.

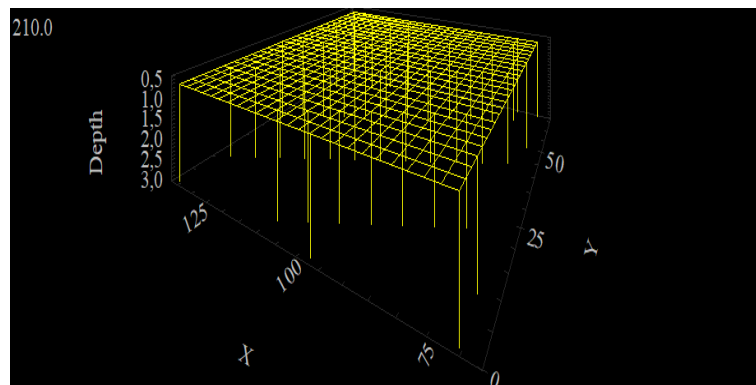


Fig. 9. Final grounding-grid geometry modeled in ETAP.

The ETAP *Ground-Grid Calculation* module was configured with the applicable body mass, ambient temperature, fault duration, and grid current. The validation results are presented in Fig. 10. Both calculated touch and step voltages remain below their corresponding tolerable limits; therefore, the design satisfies the IEEE Std 80 safety criteria for the modeled conditions.

Summary and Alert			
Result Summary			
	Calculated Volts	Tolerable Volts	
Touch	942,9	944,1	
Step	1671	3110,4	
GPR	12763	Volts	
Rg	2,326	Ohm	
Alarm & Warnings			

Fig. 10. ETAP grounding-grid safety validation.

The ETAP model was independently checked using the 6213821A283-AnexoC spreadsheet. Soil and surface-layer resistivities, grid depth, geometry, conductor data, and fault-current parameters were entered as shown in Figs. 11–14. The spreadsheet geometry and final results are shown in Figs. 12 and 15.

Datos del Suelo			
ρ	324,3	Ohm/m	(resistividad del suelo)
ρ_s	2500	Ohm/m	(resistividad superficial)
h_s	0,25	m	(Profundidad de la capa superficial)

Geometría de la malla		Ver Diagrama
Largo (X):	64 m	
Ancho (Y):	64 m	
Área:	4096 m ²	
Espacio Vertical (Ey)	2,666666667 m	} D
Espacio Horizontal (Ex)	2,666666667 m	
Conductores verticales:	25	
Conductores Horizontales:	25	
Lc:	3200 m (Longitud total de la malla)	
h:	0,50 m (Profundidad de la malla)	
		Cantidad de varillas: 44
		Largo: 2,4 m
		Con varillas en las esquinas
		LR: 105,6 m
		↓
		Lm: 3367,10 m
		Lt: 3305,60 m

Parámetros eléctricos	
Ts:	0,5 s (Tiempo de duración de la falla)
3I0:	4,18 A (3×I0 Corriente de falla)
Calcular	

Fig. 11. Spreadsheet input data: soil and grid parameters.

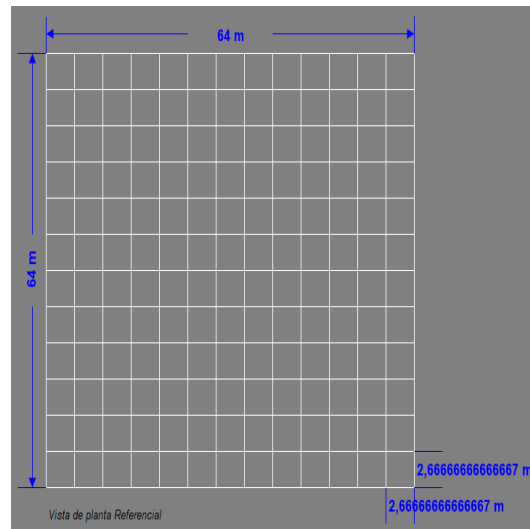


Fig. 12. Grounding-grid geometry generated by the verification spreadsheet.

Conductor de la malla

Tipo: **Cobre Comercial**

Conductividad: 97 % respecto al cobre puro

Factor α : 0,00381 @20°C [1/°C]

K0 a 0°C: 242

Tm: 1084 [°C] (Temperatura de fusión)

ρ a 20°C: 1,78 [$\mu\Omega \cdot \text{cm}$]

TCAP: 3,42 [$\text{J/cm}^3 \cdot \text{°C}$] Capacidad termica

Tipo de Union: **Soldada**

Temp Max de la Union: 450 °C

Ta: **40** °C (temperatura ambiente)

Akcmil: 0,02 kcmil

Area minima: 0,01 mm²

Diámetro mínimo: 0,0001 mm

Conductor de diseño: **20 AWG**

área: 67,42 mm²

diámetro: 0,0093 mm

*IEEE 80-2000
Sec 11.3 Tabla 1
Con temperatura de referencia 20°C*

*Características mínimas del conductor
de tierra*

Factores de paso y toque

K: -0,77 (Factor de reflexión)

Cs: 0,87 (Factor de reducción)

Peso de la persona: **70** kg

Es: 3110,37 V (Voltaje de paso Max, para el peso indicado)

Et: 944,12 V (Voltaje de toque Max)

Fig. 13. Spreadsheet input data: grounding-conductor parameters.

Calculo de corriente de malla IG

3I0: 12,35 kA (Corriente de falla)

Ts: 0,5 s (Tiempo de duracion de la falla)

X/R: **10** Relacion X/R

Df: 1,026 Factor de decremento

Sf: **0,60** Factor de división

IG: **5,42 kA** [Volver](#)

Corriente de malla

☐ Usar 3I0

☒ Introducir Valor

8,80 kA

Fig. 14. Spreadsheet input data: grid-current parameters.

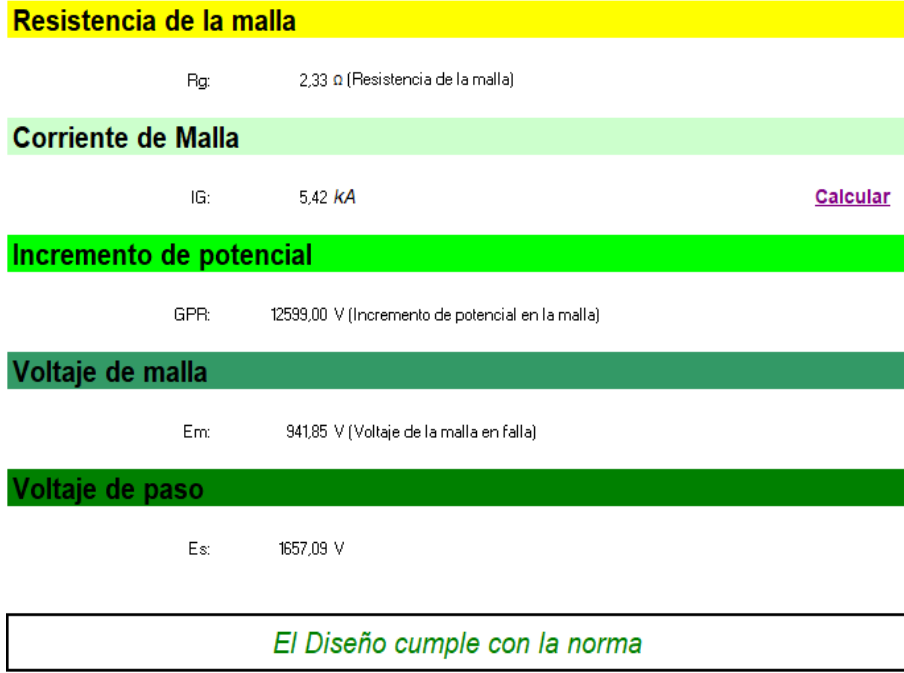


Fig. 15. Grounding-grid verification results obtained from the spreadsheet.

Table IV compares the two calculation methods. The close agreement provides an independent check of the model implementation.

TABLE IV
COMPARISON OF ETAP AND SPREADSHEET RESULTS

Calculated quantity	ETAP	Spreadsheet
Grid resistance (Ω)	2.326	2.33
Touch voltage (V)	942.9	941.85
Step voltage (V)	1671	1657.09

Both methods confirm compliance with the applicable safety limits. The small discrepancy in touch voltage is attributable primarily to treatment of the X/R ratio: ETAP uses the exact value, whereas the spreadsheet permits only discrete values of 10, 20, 30, and 40; consequently, a value of 10 was used for the spreadsheet verification.

The estimated conductor cost is USD 29,760, and the estimated rod cost is USD 493.68. The conductor dominates the material cost because the dense mesh was selected to provide adequate margin below the permissible step-voltage limit. ETAP's automatic optimization produced solutions with an impractically large number of rods; therefore, the final configuration was selected through the engineering assessment described above.

3. Surge-Arrester Selection

Surge arresters are selected for both substation voltage levels. The assumed surge impedance is 300 Ω , and the basic lightning impulse insulation level (BIL) is 350 kV. Table V summarizes the electrical design parameters.

TABLE V
SURGE-ARRESTER SELECTION PARAMETERS

Parameter	Value
Overvoltage multiplier	2.5
Surge impedance, Z_c	300 Ω
Maximum system voltage 1	750 kV
Maximum system voltage 2	500 kV
Safety factor	1.1
$R_1^{\text{Bus 9}}$	0.0533 p.u.
$X_1^{\text{Bus 9}}$	1.17 p.u.
$R_1^{\text{Bus 3}}$	0.0618 p.u.
$X_1^{\text{Bus 3}}$	1.34 p.u.
$R_0^{\text{Bus 9}}$	0.0158 p.u.
$X_0^{\text{Bus 9}}$	0.502 p.u.
$R_0^{\text{Bus 3}}$	0.0348 p.u.
$X_0^{\text{Bus 3}}$	0.662 p.u.

The positive- and zero-sequence impedances were obtained from short-circuit studies at Buses 9 and 3, as shown in Figs. 16 and 17.

SHORT-CIRCUIT REPORT

Fault at bus: Bus9

Prefault voltage = 750.000 kV
= 100.00 % of nominal bus kV (750.000 kV)
= 100.00 % of base kV (750.000 kV)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms		% Impedance on 100 MVA base			
				Va	Vb	Vc	Ia	3I0	R1	X1	R0	X0
Bus9	Total	0.00	6.599	0.00	90.88	90.29	8.146	8.146	5.33E-002	1.17E+000	1.58E-002	5.02E-001
Bus8	Bus9	0.06	2.378	0.19	90.91	90.19	2.607	2.111	8.05E-002	3.24E+000	8.34E-001	1.75E+000
Bus6	Bus9	0.04	2.112	0.08	90.92	90.23	2.492	2.389	2.50E-001	3.64E+000	7.48E-001	1.54E+000
Bus3	Bus9	27.40	2.109	61.33	100.00	60.58	3.099	4.089 *	1.69E-001	3.65E+000	2.93E-002	1.00E+000

Fig. 16. Short-circuit study results at Bus 9.

SHORT-CIRCUIT REPORT

Fault at bus: **Bus3**

Prefault voltage = 500.000 kV
 = 100.00 % of nominal bus kV (500.000 kV)
 = 100.00 % of base kV (500.000 kV)

Contribution		3-Phase Fault		Line-To-Ground Fault					Positive & Zero Sequence Impedances Looking into "From Bus"			
From Bus ID	To Bus ID	% V From Bus	kA Symm. rms	% Voltage at From Bus			kA Symm. rms Ia	3I0	% Impedance on 100 MVA base			
				Va	Vb	Vc			R1	X1	R0	X0
Bus3	Total	0.00	8.611	0.00	91.41	91.70	10.358	10.358	6.18E-002	1.34E+000	3.48E-002	6.62E-001
Bus9	Bus3	36.84	4.254	65.62	65.93	100.00	3.411	0.000	1.07E-001	2.71E+000		
Gen3	Bus3	111.11	4.358	111.11	111.11	111.11	6.947	10.358	1.39E-001	2.65E+000	3.48E-002	6.62E-001

Fig. 17. Short-circuit study results at Bus 3.

The calculation begins at the substation side that produces the higher fault current, namely Bus 3. The ratio

$$\frac{R_1}{X_1} \quad (2)$$

is first evaluated to select the appropriate coefficient curve in Fig. 18. The coordinates used on the selected curve are

$$\frac{R_0}{X_1}, \quad \frac{X_0}{X_1}. \quad (3)$$

The region containing the calculated point defines the grounding or correction factor K_e .

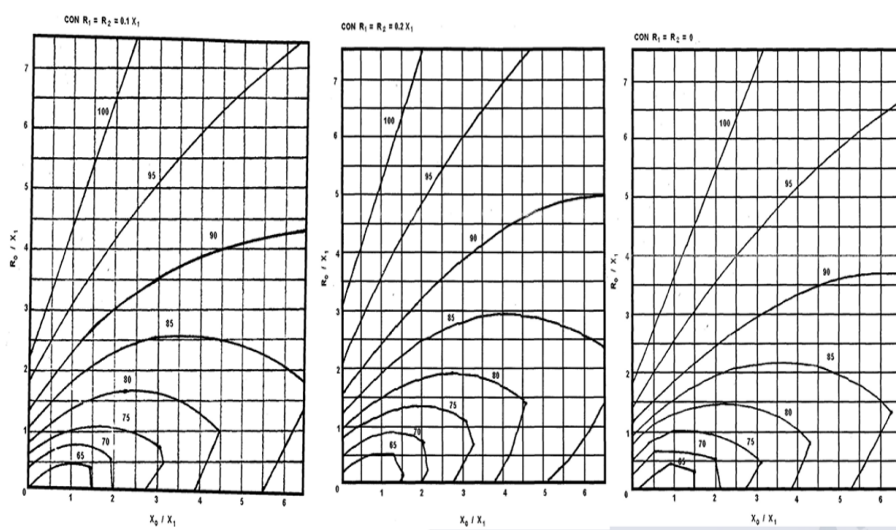


Fig. 18. Coefficient curves as a function of sequence-impedance ratios.

The arrester rated voltage, required discharge-current capability, and maximum prospec-

tive overvoltage are calculated as

$$V_r = K_e V_{\max}, \quad (4)$$

$$I_A = \frac{2 \text{ BIL}}{Z_c}, \quad (5)$$

$$E_{\max} = V_{\max} \frac{\sqrt{2}}{\sqrt{3}} K_{\text{OV}} K_S, \quad (6)$$

where $K_{\text{OV}} = 2.5$ is the overvoltage multiplier and $K_S = 1.1$ is the design safety factor.

3.1 Arrester for Bus 3

Application of (2) and (3) gives $R_1/X_1 = 0.0455$, which is sufficiently close to zero to use the $R_1 = 0$ curve. The corresponding coordinates are $X_0/X_1 = 0.4940$ and $R_0/X_1 = 0.0260$, locating the operating point in the 65% region. Therefore, $K_e = 0.65$.

For $V_{\max} = 500$ kV,

$$V_r = 0.65(500) = 325 \text{ kV},$$

$$I_A = \frac{2(350)}{300} = 2.333 \text{ kA},$$

$$E_{\max} = 500 \frac{\sqrt{2}}{\sqrt{3}} (2.5)(1.1) = 1122.683 \text{ kV}.$$

The arrester is selected from the ABB high-voltage surge-arrester catalog [5]. The applicable catalog ratings are shown in Fig. 19. For lightning impulses, the residual-voltage data associated with the 8/20 μs current waveform are used.

Max. system voltage U_s kV _{rms}	Rated voltage U_r kV _{rms}	Max. continuous operating voltage ¹⁾		TOV capability ²⁾		Max. residual voltage with current wave							
		as per IEC U_c kV _{rms}	as per ANSI/IEEE MCOV kV _{rms}	1 s kV _{rms}	10 s kV _{rms}	30/60 μs			8/20 μs				
						1 kA kV _{peak}	2 kA kV _{peak}	3 kA kV _{peak}	5 kA kV _{peak}	10 kA kV _{peak}	20 kA kV _{peak}	40 kA kV _{peak}	
300	228	182	182	251	239	445	461	473	493	519	568	623	
	240	191	191	265	252	468	485	497	519	546	598	656	
	258	191	209	285	270	504	522	535	558	587	643	705	
	264	191	212	291	277	515	534	547	571	601	658	721	
	276	191	220	304	289	539	558	572	597	628	688	754	
362	258	206	209	285	270	504	522	535	558	587	643	705	
	264	211	212	291	277	515	534	547	571	601	658	721	
	276	221	221	304	289	539	558	572	597	628	688	754	
	288	230	230	318	302	562	582	597	623	656	718	787	
420	330	264	267	364	346	644	667	684	714	751	823	901	
	336	267	272	371	352	656	679	696	727	765	838	918	
	342	267	277	377	359	667	691	709	740	779	852	934	
	360	267	291	397	378	702	728	746	779	819	897	983	
	378	267	306	417	396	737	764	783	817	860	942	1037	
	390	267	315	430	409	761	788	808	843	888	972	1070	
	396	267	318	437	415	773	800	820	856	901	987	1086	
	420	336	336	464	441	819	849	870	908	956	1051	1152	
550	444	349	353	490	466	866	897	920	960	1015	1111	1217	

Fig. 19. Candidate surge-arrester ratings for Bus 3.

The nearest catalog maximum system voltage above 500 kV is 550 kV. The arrester must also exceed the calculated 2.333 kA discharge-current requirement and withstand the calculated maximum overvoltage of 1122.683 kV. A PEXLIM P-Y metal-oxide arrester with a 40 kA rating and a listed peak withstand level of 1217 kV is selected. Its principal characteristics are reproduced in Fig. 20.

Brief performance data	
Arrester classification as per IEC 60099-4 Ed 3.0	Station; SH
Arrester classification as per IEEE Std C62.11-2012	Station
System voltages (U_d)	300 - 550 kV
Rated voltages (U_r)	228 - 444 kV
Nominal discharge current (IEC)	20 kA _{peak}
Lightning impulse classifying current (ANSI/IEEE)	10/15 kA _{peak}
Charge, energy and current withstand:	
Repetitive charge transfer rating, Q_{rs} (IEC)	3.2 C
Thermal energy rating, W_{th} (IEC)	11 kJ/kV (U_d)
Single impulse energy capability (2 ms to 4 ms impulse)	7.0 kJ/kV (U_d)
Discharge current withstand strength:	
High current 4/10 μ s	100 kA _{peak}
Low current 2000 μ s, (based on Q_{rs})	1 600 A _{peak}
Energy class as per IEEE standard (switching surge energy rating)	G
Single-impulse withstand rating as per IEEE standard	3.2 C
Repetitive charge transfer test value - sample tests on all manufactured block batches	4.0 C
Short-circuit/Pressure relief capability	65 kA _{rms(sym)}
Mechanical strength:	
Specified long-term load (SLL)	6000 Nm
Specified short-term load (SSL)	9000 Nm
Service conditions:	
Ambient temperature	-50 °C to +45 °C
Design altitude	max. 1000 m
Frequency	15 - 62 Hz
Line discharge class (as per IEC60099-4, Ed. 2.2)	Class 4



Fig. 20. Selected PEXLIM P-Y zinc-oxide surge arrester.

3.2 Arrester for Bus 9

The same sequence-impedance procedure places the Bus 9 operating point on the same coefficient curve; hence, $K_e = 0.65$. For $V_{\max} = 750$ kV,

$$\begin{aligned}
 V_r &= 0.65(750) = 487.5 \text{ kV}, \\
 I_A &= 2.333 \text{ kA}, \\
 E_{\max} &= 750 \frac{\sqrt{2}}{\sqrt{3}} (2.5)(1.1) = 1684.024 \text{ kV}.
 \end{aligned}$$

The current value uses the same 350 kV BIL because a separate BIL was not provided for this voltage level.

The same manufacturer catalog is used for the high-voltage-side selection. Figure 21 lists the available alternatives.

Max. system voltage U_s	Rated voltage U_r	Max. continuous operating voltage ¹⁾		TOV capability ²⁾		Max. residual voltage with current wave							
		as per IEC	as per ANSI/IEEE	1 s	10 s	30/50 μ s							
		kV _{max}	kV _{max}			1 kA	2 kA	3 kA	5 kA	10 kA	20 kA	40 kA	
245	180	144	144	199	199	346	346	346	381	396	406	445	
	192	154	154	212	201	369	369	369	406	423	437	467	
	198	158	160	218	206	382	382	382	419	436	451	481	
	210	166	170	232	220	404	416	433	444	460	469	513	
	216	168	174	238	227	416	427	438	457	476	476	514	
	219	168	177	242	230	421	433	441	463	482	482	519	
	222	168	179	245	233	427	439	447	469	489	489	526	
	225	168	180	250	236	438	451	459	482	502	502	539	
	216	173	174	238	227	416	427	438	457	476	476	514	
	225	182	182	252	239	438	451	459	482	502	502	539	
	240	191	191	265	252	461	475	484	507	528	528	567	
300	225	181	200	255	271	459	510	520	540	568	574	617	
	240	191	212	265	277	468	522	532	558	581	581	623	
	270	191	220	295	290	511	546	556	583	608	608	651	
	265	206	209	285	271	496	510	520	545	568	574	617	
	275	211	212	291	277	508	522	532	558	581	581	623	
	276	221	221	305	290	531	545	555	583	608	608	651	
	276	230	230	318	302	554	569	580	609	634	634	676	
	300	244	247	354	347	634	652	665	697	726	735	784	
	300	257	272	371	363	646	664	677	710	740	749	799	
	342	257	277	378	368	657	675	689	723	753	763	813	
	342	267	291	398	378	692	712	725	761	792	805	851	
	375	267	301	411	391	719	735	749	786	819	824	872	
362	375	267	306	418	397	726	747	761	799	832	839	887	
	381	267	306	421	400	732	753	767	805	839	846	895	
	380	267	315	431	410	750	771	785	824	858	867	915	
	395	267	318	437	416	761	783	798	837	872	877	925	
	420	267	336	467	441	807	832	848	888	924	931	981	
	395	317	318	437	416	761	783	798	837	872	877	925	
	420	336	336	464	441	807	832	848	888	924	931	981	
	444	348	353	491	467	853	878	894	934	977	1000	1050	
	465	470	470	550	518	1104	1167	1183	1247	1309	1402	1456	
	412	490	490	676	643	1180	1214	1237	1298	1361	1459	1497	
	520	490	490	690	660	1205	1238	1261	1323	1378	1488	1516	

Fig. 21. Candidate surge-arrester ratings for Bus 9.

The available catalog extract does not provide all data needed to verify the selection directly against the calculated maximum voltage and discharge current. The preliminary selection therefore uses the catalog unit with a maximum system voltage of 800 kV, which exceeds the specified 750 kV system voltage, and applies the 8/20 μ s lightning-current ratings. The selected TEXLIM T-C metal-oxide arrester is shown in Fig. 22. Final procurement shall be subject to manufacturer confirmation of the required protective level, temporary-overvoltage capability, energy duty, and nominal discharge-current rating.

Brief performance data

Arrester classification as per IEC 60099-4 Ed 3.0	Station; SH
Arrester classification as per IEEE Std C62.11-2012	Station
System voltages (U_s)	245 - 800 kV
Rated voltages (U_r)	180 - 624 kV
Nominal discharge current (IEC)	20 kA _{peak}
Lightning impulse classifying current (ANSI/IEEE)	10/15/20kA _{peak}
Charge, energy and current withstand:	
Repetitive charge transfer rating, Q_{rs} (IEC)	5.2 C
Thermal energy rating, W_{th} (IEC)	15 kJ/kV (U_r)
Single impulse energy capability (2 ms to 4 ms impulse)	11 kJ/kV (U_r)
Discharge current withstand strength:	
High current 4/10 μ s	150 kA _{peak}
Low current 2000 μ s, (based on Q_{rs})	2600 A _{peak}
Energy class as per IEEE standard (switching surge energy rating)	J
Single-impulse withstand rating as per IEEE standard	5.2 C
Repetitive charge transfer test value - sample tests on all manufactured block batches	6.2 C
Short-circuit/Pressure relief capability	80 kA _{rms(sym)}
Mechanical strength:	
Specified long-term load (SLL)	21000 Nm
Specified short-term load (SSL)	40000 Nm
Service conditions:	
Ambient temperature	-50 °C to +45 °C
Design altitude	max. 1000 m
Frequency	15 - 62 Hz
Line discharge class (as per IEC60099-4, Ed. 2.2)	Class 5



Fig. 22. Preliminary TEXLIM T-C zinc-oxide surge-arrester selection.

4. Direct-Lightning Shielding

The lightning-shielding design for Substation 3 uses the following parameters:

- substation BIL: 350 kV;
- air-terminal or shield-wire support height, H : 18.5 m; and

- protected-equipment height, A : 5.5 m.

IEEE Std 998 describes alternative methods for designing direct-stroke shielding systems [6]. This analysis applies the electrogeometric model and rolling-sphere construction. The minimum shielding current is

$$I_s = \frac{2.2 \text{ BIL}}{Z_c}, \quad (7)$$

where I_s is expressed in kiloamperes, BIL is the basic lightning impulse insulation level, and Z_c is the surge impedance. The corresponding striking distance is

$$S = 8kI_s^{0.65}, \quad (8)$$

where S is in metres, $k = 1.0$ for shield wires, and $k = 1.2$ for vertical air terminals.

4.1 Shielding with Vertical Air Terminals

Figure 23 defines the rolling-sphere geometry for a single vertical air terminal. Here, H is the terminal height, A is the protected-equipment height, and T is the horizontal clearance from the terminal to the protected object.

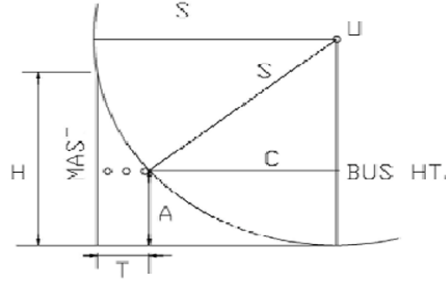


Fig. 23. Rolling-sphere geometry for one vertical air terminal.

For the specified insulation level and $Z_c = 300 \, \Omega$,

$$I_s = \frac{2.2(350)}{300} = 2.567 \text{ kA},$$

$$S = 8(1.2)(2.567)^{0.65} = 17.72 \text{ m}.$$

At an equipment height of 5.5 m, the rolling-sphere chord is

$$C = \sqrt{S^2 - (S - A)^2}$$

$$= \sqrt{17.72^2 - (17.72 - 5.5)^2} = 12.832 \text{ m},$$

and the corresponding horizontal clearance is

$$T = S - C = 4.888 \text{ m}.$$

The final arrangement uses four vertical terminals in a square configuration, as shown in Fig. 24. The geometry must maintain the rolling sphere above the 5.5 m protection plane.

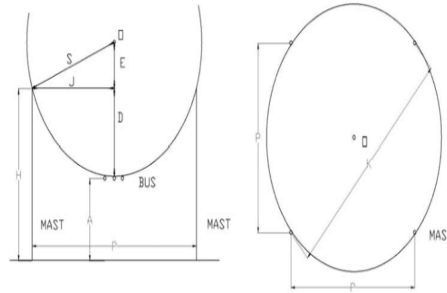


Fig. 24. Rolling-sphere geometry for four vertical air terminals.

The relevant dimensions are

$$\begin{aligned} D &= H - A = 18.5 - 5.5 = 13.0 \text{ m}, \\ E &= S - D = 17.72 - 13.0 = 4.72 \text{ m}, \\ J &= \sqrt{S^2 - E^2} = \sqrt{17.72^2 - 4.72^2} = 17.08 \text{ m}, \\ K &= 2J = 34.16 \text{ m}. \end{aligned}$$

For a square arrangement, the diagonal relation is $P^2 + P^2 = K^2$; therefore,

$$P = \frac{K}{\sqrt{2}} = \frac{34.16}{\sqrt{2}} = 24.154 \text{ m.}$$

Accordingly, four 18.5 m vertical air terminals arranged on a square with a maximum side length of 24.154 m provide shielding for equipment up to 5.5 m high under the stated assumptions.

4.2 Shielding with Overhead Shield Wires

Figure 25 defines the rolling-sphere geometry for one overhead shield wire.

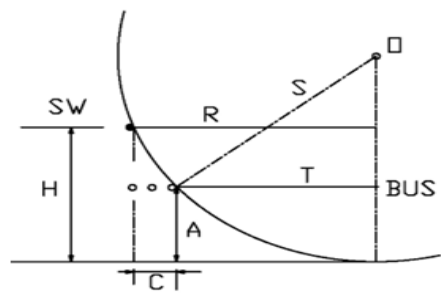


Fig. 25. Rolling-sphere geometry for one overhead shield wire.

The minimum shielding current remains $I_s = 2.567$ kA. With $k = 1.0$ for a shield wire,

the striking distance becomes

$$S = 8(1.0)(2.567)^{0.65} = 14.7644 \text{ m.}$$

The horizontal coordinates in Fig. 25 are

$$\begin{aligned} R &= \sqrt{S^2 - (H - S)^2} \\ &= \sqrt{14.7644^2 - (18.5 - 14.7644)^2} = 14.284 \text{ m,} \\ T &= \sqrt{S^2 - (S - A)^2} \\ &= \sqrt{14.7644^2 - (14.7644 - 5.5)^2} = 11.496 \text{ m,} \\ C &= R - T = 14.284 - 11.496 = 2.788 \text{ m.} \end{aligned}$$

Thus, a single shield wire provides a horizontal protected distance of approximately 2.788 m beyond the indicated geometric boundary at the equipment elevation.

The preferred design uses two parallel shield wires so that the rolling sphere remains above the protected plane, as illustrated in Fig. 26. The required maximum wire separation is denoted by X .

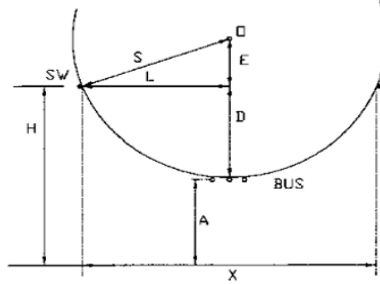


Fig. 26. Rolling-sphere geometry for two overhead shield wires.

For this configuration,

$$\begin{aligned} D &= H - A = 18.5 - 5.5 = 13.0 \text{ m,} \\ E &= S - D = 14.7644 - 13.0 = 1.7644 \text{ m,} \\ L &= \sqrt{S^2 - E^2} \\ &= \sqrt{14.7644^2 - 1.7644^2} = 14.658 \text{ m,} \\ X &= 2L = 29.317 \text{ m.} \end{aligned}$$

Therefore, the two overhead shield wires shall be separated by no more than 29.317 m to protect equipment up to 5.5 m high.

4.3 Results and Engineering Interpretation

The electrogeometric analysis yields a maximum square spacing of 24.154 m for four vertical air terminals and a maximum separation of 29.317 m for two overhead shield wires. In both cases, the design criterion is to prevent the rolling sphere from intersecting the equipment protection plane. These values are maximum geometric spacings for the stated BIL, surge impedance, and installation heights; detailed design should additionally verify conductor sag, structure geometry, edge effects, equipment clearances, and the complete substation plan.

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